Two Operator Problems for Generalized Sugeno Integrals: Explicit Later Answers

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Status: literature already answered (both stated open problems, full).

Original questions

Boczek and Kaluszka introduce $\star$-associated functions by requiring

$$\inf_{x\in A}\big(f(x)\star g(x)\big) = \left(\inf_{x\in A} f(x)\right)\star\left(\inf_{x\in A} g(x)\right)$$

for every nonempty measurable A. Their **Open Problem 1**, on source PDF page 3, asks whether an operator $\star \neq +$ can make this property equivalent to comonotonicity [1].

Their **Open Problem 2**, on source PDF page 10, equation (20), asks whether there is a pair $(\nabla, \triangle) \neq (\max, \min)$ for which the generalized lower and upper Sugeno integrals agree for every measurable function:

$$\inf_{t\in Y} \nabla(t, \mu(D\cap\{f>t\})) = \sup_{t\in Y} \triangle(t, \mu(D\cap\{f\geq t\}))\,.$$

Explicit later answers

Hutník and Pócs explicitly identify both questions as the two open problems of Boczek–Kaluszka and state that they answer both positively [2].

For Problem 1, their Theorem 2.7 (journal page 274; supporting PDF page 7) states that if $\star : Y^2 \to Y$ is strictly monotone and right-continuous, then $\star$-associatedness is equivalent to comonotonicity on every measurable space. This is a broad affirmative answer. For example, whenever it maps the chosen interval Y into itself, the arithmetic mean $a \star b = (a+b)/2$ is a non-addition instance.

For Problem 2, their Theorem 3.1 (journal page 276; supporting PDF page 9) states that every order-preserving continuous $\varphi : [0,\infty] \to [0,\infty]$ gives a valid pair

$$x\nabla y = \varphi(\max\{x,y\}), \qquad x\triangle y = \varphi(\min\{x,y\}).$$

The proof uses preservation of infima and suprema by φ and the usual lower/upper representation of the Sugeno integral. Choosing a nonidentity φ, for instance $\varphi(s) = s/2$ with $\varphi(\infty) = \infty$, gives the requested pair distinct from $(\max, \min)$.

The supporting authors knew that they were answering the source questions; this is therefore an explicit literature answer, not an inferred reformulation.

Scope and search evidence

Both existential questions as literally stated are completely resolved. Hutník–Pócs Remark 3.2 notes that the stronger classification of *all* operator pairs producing coincident generalized integrals remains unknown; that classification is not part of Boczek–Kaluszka Open Problem 2.

Exact-title, exact-question, arXiv-id, author, and citation searches located the 2018 Kybernetika article, whose abstract and introduction expressly say that it answers the two problems. The theorem statements above were checked in the journal PDF against the original statements in arXiv:1506.08567.

References

[1] M. Boczek and M. Kaluszka, *On the Minkowski–Hölder type inequalities for generalized Sugeno integrals with an application*, Kybernetika **52** (2016), no. 3, 329–347; arXiv:1506.08567.

[2] O. Hutník and J. Pócs, *On $\star$-associated comonotone functions*, Kybernetika **54** (2018), no. 2, 268–278; doi:10.14736/kyb-2018-2-0268.